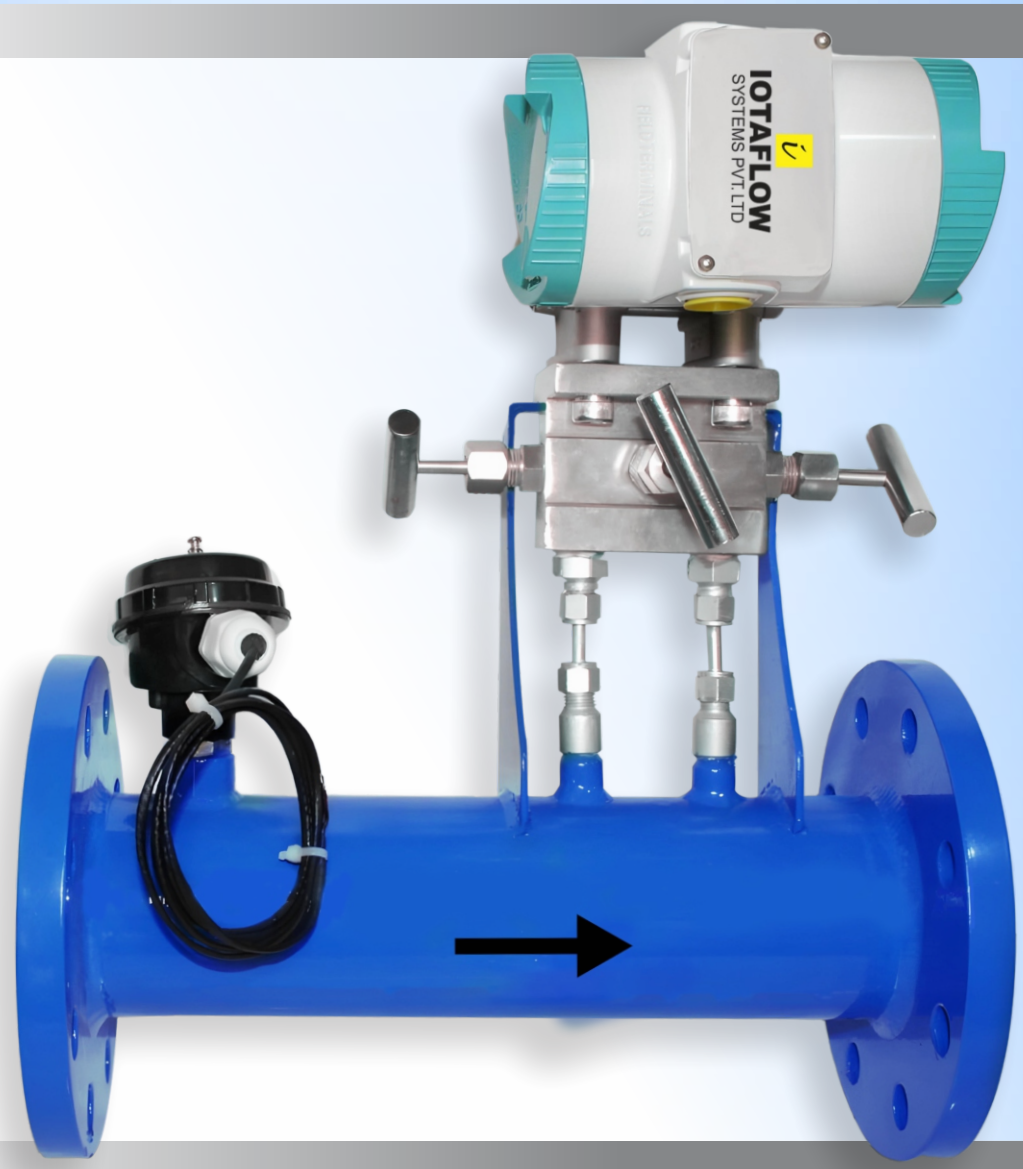
" if you want to manage it Measure it "

IOTAFLOW
SYSTEMS PVT. LTD



CONICAL SHAPED FLOW METER

www.iotaflow.com

OVERVIEW

The IOTAFLOW **Conical Shaped Flow Meter (CSF)** technology provides precise flow measurements across a broad range of Reynolds numbers, in a wide range of environments, and for a large variety of fluids. It uses the concept of energy conservation in fluid flow through a pipe, the same fundamental mechanism as other differential pressure-typeflow meters, to function. **CSF's** exceptional performance qualities are the outcome of its distinctive design. Inside the tube,it has a cone that is positioned in the middle. A zone of reduced pressure is produced immediately downstream of the cone as a result of the cone's interaction with the fluid flow, which alters the flow profile of the fluid. Two pressure sensing taps can be used to monitor the pressure differential between the static line pressure and the low pressure produced downstream of the cone. Two taps are positioned: one in the downstream face of the cone itself, the other a little upstream of the cone. The fluid flow rate is then calculated using a derivation of the Bernoulli equation that incorporates the pressure difference

FEATURES

Design Benefits

The Conical Shaped Flow Meter reshapes the velocity profile upstream due to the contoured form and placement of the hanging cone. Even under extremely turbulent flow conditions, the flow profile "flattens" toward the form of a well-developed profile as it approaches the cone. No of the flow conditions upstream of the meter, the cone's central location in the line optimizes the flow velocity at the moment of measurement, ensuring extremely precise, dependable flow measurement.

Outstanding Performance in Difficult Applications

The Conical Shaped Flow Meter from IOTAFLOW is a cutting-edge technology that raises the bar for differential pressure flow monitoring. This innovative flow meter regularly surpasses conventional DP devices and other flow technologies. It is designed for mild to demanding working situations and for a broad variety of fluids. Better accuracy and repeatability, a larger range, flexible installation, and less maintenance are all features of CSF.

Performs Own Flow Conditioning

The shape and placement of the cone in respect to the measurement ports is what contributes to the Conical Shaped Flow Meter's improved performance. This prevents any centralized flow disruptions, enabling the CSF meter to function as its own flow conditioner. Readings are always accurate and trustworthy, even in low pressure flow scenarios, because to the thoroughly mixed and conditioned flow, which produces a low amplitude, high frequency signal with minimum "signal noise".

Installation Freedom

Installation Freedom stems from the Conical Shaped Flow Meter's ability to condition the flow before measuring. The CSF meter doesn't need the upstream or downstream straight pipe lines that many other flow meters use since it can detect disturbed flow properly. The CSF meter may be mounted almost anywhere in a pipe system thanks to this crucial feature, and it is simple to retrofit into an existing piping configuration. Cost savings that are large might be the outcome. Additionally, it implies that the CSF meter may fit in places where traditional flow meters cannot because of space or weight restrictions.

Negligible Operational costs

The long-term functioning of the Conical Shaped Flow Meter flow meter is guaranteed. There are **no moving parts** to repair and replace. Furthermore, the flow is directed by the conical contoured shape rather than coming into contact with an abrupt surface. As a result, fluid is directed away from the beta edge by a boundary layer that develops along the cone. Compared to other flow meter methods, the meter's calibration is accurate for a considerably longer period since the beta doesn't vary.

Customization Fulfills a Variety of Needs

The Customization Offers Outstanding Sizing Flexibility to Meet wide range of requirements. It is sized for line diameters ranging from 1/2" to more than 120". There is also a wide range of building supplies accessible.

Highly Accurate

The Conical Shaped Flow Meter's capacity to deliver repeated accuracy of up to +/- 1.0% of rate under even the most challenging flow circumstances is its primary advantage. It has been established that the CSF meter is reliable throughout a broad range, from very low to extremely high Reynolds numbers. The CSF meter offers the precision and dependability that conventional instruments can only supply when used in a laboratory, whether monitoring whirling fluids or low pressure flows. When compared to other dP technologies, the CSF meter also has a reduced head loss.

- Ideal for tough applications
- High accuracy
- High repeatability
- Self conditioning
- Low signal noise
- Minimum straight pipe requirements
- Broad rangeability
- Low headloss
- Clean or dirty liquids, wet gases, slurries
- Virtually no maintenance

Standard Accuracy	From ± 1.0% of actual flow (certain fluids and Reynolds number applications require special calibrations to achieve this value)
Repeatability	±0.5% or better
Flow Ranges	10:1 and greater
Standard Beta Ratios	0.45 through 0.85, special betas available
Head Loss	Varies with beta ratio and DP
Installation Piping Requirements	Typically 0-3 diameters upstream and 0-1 diameters downstream of the cone are required, depending on fittings or valves in the adjacent Pipeline.
Materials of Construction Include	Duplex, 304, or 316 stainless steel, Hastelloy C-276, carbon steels Other materials on request
Line Sizes	0.5” to 120” or larger
End Fittings	Flanged, threaded, hub or weld-end standard Others on request
Configurations	Precision flow tube, Insertion and wafer-type • Calibrated for customer application • ASME 1967 construction standard • IAPWS 1995 standards ISO 5167-5 • ISO 9001:2015 certified quality management system

SELECTION SHEET

IOTAFLOW make Conical Shaped Flow Meter												
	Conical Shaped Flow - CSF	XXXX	X	X	X	X	X	X	X	X	X	X
Size	DN25	25										
	DN32	32										
	DN40	40										
	DN50	50										
	DN65	65										
	DN80	80										
	DN100	100										
	DN125	125										
	DN150	150										
	DN200	200										
	DN250	250										
	DN300	300										
	DN400	400										
	DN500	500										
	DN600	600										
	DN700	700										
	DN800	800										
	DN900	900										
	DN1000	1000										
	DN1200	1200										
	DN1500	1500										
Type	Insertion	I	Insertion									
	Inline	L	Inline									
Body	SS316	F		SS 316								
	CS	D		Carbon Steel								
	SS304	S		SS 304								
	Not Applicable	O		NA								
Cone Material	CS	D			Carbon Steel							
	SS316	F			SS 316							
	SS 304	S			SS 304							
Flange Material	SS316	F				SS 316						
	CS	D				Carbon Steel						
	SS304	S				SS 304						
	Not Applicable	O				NA						
Pressure Rating	Pressure <10bar	1					up to 10 bar					
	Pressure16bar	2					up to 16 bar					
	Pressure25bar	3					up to 25 bar					
	Pressure40bar	4					up to 40 bar					
	Pressure64bar	5					up to 64 Bar					
	Pressure>64bar	6					> 64 bar					
Temperature Suit	Temperature <500	H						<500 deg C				
	Temperature <250	N						<250 deg C				
Service	Service Gas	G							Gas			
	Service Air	A							Air			
	Service Steam	S							Steam			
	Service Water	W							Water			
	Service Oil	O							Oil			
DPT	Provided	D								Diff Pre Tx		
	Skip	O								N.A.		
3 Way Manifold	Provided	M									3 way Manifold SS	
	Skip	O									N.A.	
PTC	Skip	O										N.A.
	Provided	PTC										Pressure & Temp. Correction
Digital Flow Rate Indicator Totalizer FI-106	Skip	O										
	Provided	FI_106										

Contact for Support

Contact your IOTAFLOW'S representative right away for an assessment of your flow application or to learn more about our other flow meter products.

Contact Us

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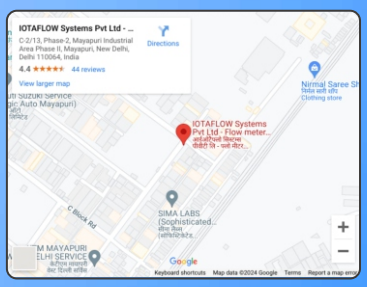

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